

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient is diagnosed with community-acquired pneumonia , which is caused by bacteria . Bacteria , like Streptococcus pneumoniae , have specific antigens on their cell wall , such as peptidoglycan , which can trigger an immune response . The Th1 response is often associated with bacterial infections . </think><answer> Peptidoglycan is its major antigen that induces an immune response .</answer>

